

Supplementary Materials for “RAPT: A Transformer-based Approach for Robust Symbolic Score-to-Performance Alignment”

This document contains supplementary materials for the paper titled “RAPT: A Transformer-based Approach for Robust Symbolic Score-to-Performance Alignment” submitted to ISMIR 2026.

1 Model Hyperparameters

For reproducibility, the full list of hyperparameters of the RAPT architecture and training pipeline are shown in Table 1, and the Optuna-tuned inference hyperparameters are shown in Table 2.

Symbol	Description	Value
F	Temporal resolution (quantization)	50 steps/s
W	Context window	37 (0.74 s)
d	Embedding dimension	256
L	Transformer layers	3
H	Attention heads	4
d_{ff}	Feed-forward dimension	512
K	Positional conv kernel	31
τ	InfoNCE temperature	0.07
p_{drop}	Note dropout probability	0.1
R	Sakoe-Chiba radius	50
θ_{keep}	Fusion IoU threshold	tuned per dataset
Δ_{max}	Max gap interpolation	1.5 s
Max epochs	Training budget upper bound	200 (15 for Batik)
Patience	Early-stopping patience	15 (999 for Batik)
Split seed	Train/val/test partition seed	1337
Weight seeds	Per-ensemble-member init seeds	{1001, 2002, 3003, 4004}

Table 1: Architectural and pipeline hyperparameters.

Configuration	α	β	θ_{keep}
Vienna 4x22	0.137	0.770	0.892
Batik plays Mozart	0.231	0.499	0.832
(n)ASAP	0.834	0.944	0.900
Joint training: ASAP+Vienna	0.076	0.695	0.852
Joint training: ASAP+Batik	0.332	0.804	0.868

Table 2: Optuna-tuned hyperparameters used for evaluation. The first three rows are in-domain configurations; the last two are joint training configurations used for cross-corpus evaluation.

2 Detailed Experimental Results

This section contains supplementary results and plots, to those results and tables presented in Section 4 in the paper.

	ASAP	Batik	Vienna
Nakamura	0.9706 $\pm$ 0.0292	0.9953 $\pm$ 0.0032	0.9841 $\pm$ 0.0199
DualDTW	0.9799 $\pm$ 0.0222	0.9907 $\pm$ 0.0064	0.9887 $\pm$ 0.0064
GlueNote	0.9688 $\pm$ 0.0295	0.9800 $\pm$ 0.0140	0.9763 $\pm$ 0.0097
RAPT (Ours)	0.9780 $\pm$ 0.0226	0.9884 $\pm$ 0.0071	0.9920 $\pm$ 0.0115

Table 3: Comparison of RAPT against SOTA baselines on clean datasets (F1 Score: Mean $\pm$ Std).

Method	Vienna Δ	Batik Δ	ASAP Δ
Nakamura	-.225	-.003	<u>-.001</u>
DualDTW	-.220	<u>-.001</u>	-.001
GlueNote	<u>-.221</u>	-.002	-.002
RAPT (Ours)	-.223	-.001	.000
RAPT vs Baselines	$p > .05^\dagger$	$p > .05^\dagger$	$p < .001$

† Low test-set size (descriptive only).

Table 4: Degradation summary ($\Delta = \text{F1}_{\text{corrupt}} - \text{F1}_{\text{clean}}$). Negative values indicate performance loss under corruption. Wilcoxon signed-rank tests (Holm-Bonferroni corrected) compare RAPT against all three baselines.

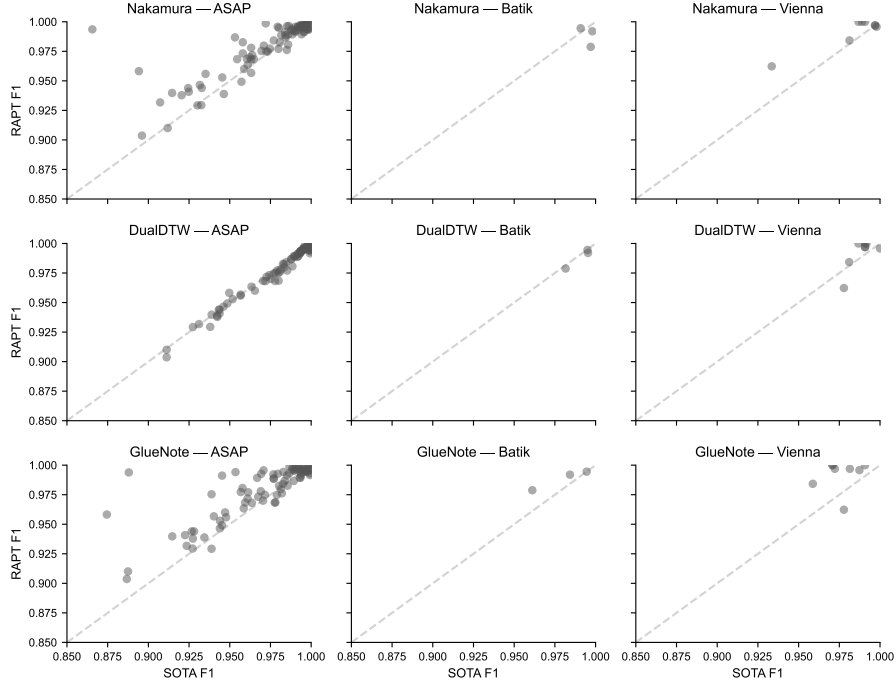


Figure 1: F1 scores of individual pieces comparing SOTA and our model RAPT

3 Ablation Study

This section contains supplementary details to the ablation study presented in Section 4.4 in the paper.

Ablation	Vienna	Batik	ASAP
RAPT (Canonical)	.992	.988	.978
Pure Transformer	.746	.891	.815
88-dim input	.991	.988	.977
No gap interpolation	.989	.989	.978
Mean pooling	.991	.989	.977
Max pooling	.992	.989	.977
IoU threshold=0	—	.978	—
Single seed	.991	.988	.975

Table 5: Ablation study on aggregate F1 scores (clean condition). The **Canonical RAPT** model serves as the baseline.

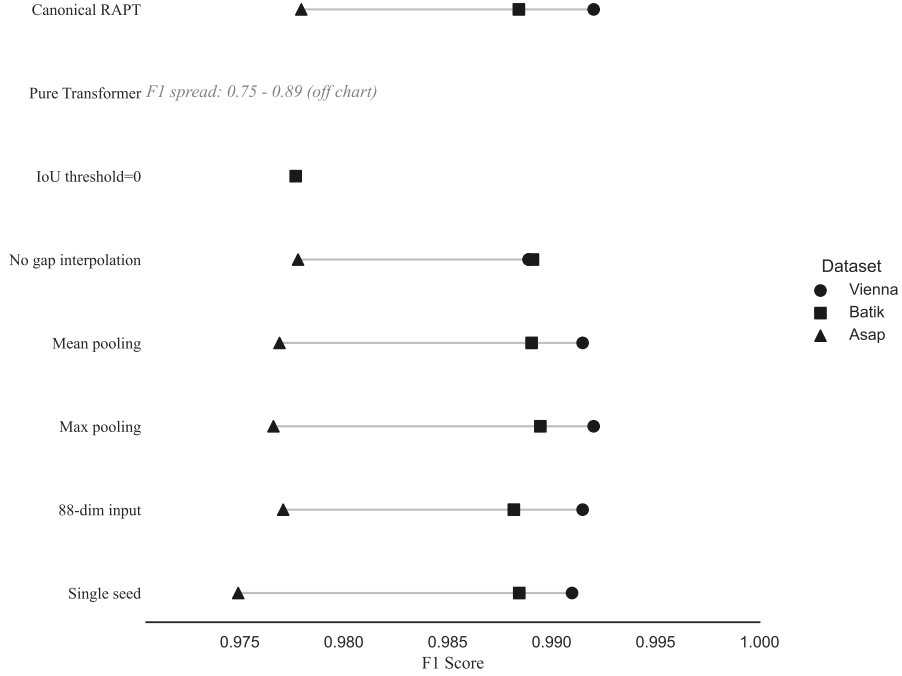


Figure 2: F1 scores of various configurations across datasets

4 Cross-Corpus Transfer

Method	ASAP+Batik→Vienna ($n = 88$)	ASAP+Vienna→Batik ($n = 36$)
Nakamura	0.976	0.981
GlueNote	0.974	0.973
DualDTW	0.987	0.987
RAPT	0.986	0.986

Table 6: Cross-corpus alignment F1: each ensemble is trained on two corpora and evaluated on the full held-out third corpus.

5 Compass Alignment Examples

Piece	RAPT	DualDTW	Nakamura	GlueNote
Chopin op. 38, p. 08 (V)	1.000	0.992	0.978	0.971
Chopin op. 38, p. 11 (V)	1.000	0.987	0.971	0.974
Schubert D.899/1 (A)	0.957	0.950	0.937	0.875
Haydn Sonata 31-1 (A)	0.999	0.997	0.999	0.989

Table 7: Per-piece F1 scores on a sample of pieces from Vienna 4x22 (V) and ASAP (A) highlighting where RAPT’s contextual embeddings successfully disambiguate challenging regions.

6 Statistical Testing

This section contains the detailed results of the statistical tests for comparing the performance of the different methods.

Dataset	Comparison	Test	Statistic	p-value
ASAP	ANOVA (All)	F-test	3.4539	1.68e-02 *
	RAPT vs Nakamura	Wilcoxon/Mann-Whitney	671.5000	1.15e-06 ***
	RAPT vs DualDTW	Wilcoxon/Mann-Whitney	540.0000	4.59e-08 ***
	RAPT vs GlueNote	Wilcoxon/Mann-Whitney	545.5000	5.42e-08 ***
Batik	ANOVA (All)	F-test	1.1207	3.96e-01 ns
	RAPT vs Nakamura	T-test	-1.0817	3.92e-01 ns
	RAPT vs DualDTW	T-test	-2.5705	1.24e-01 ns
	RAPT vs GlueNote	T-test	1.6551	2.40e-01 ns
Vienna	ANOVA (All)	F-test	1.9218	1.49e-01 ns
	RAPT vs Nakamura	T-test	2.1908	6.46e-02 ns
	RAPT vs DualDTW	T-test	1.0215	3.41e-01 ns
	RAPT vs GlueNote	T-test	2.9393	2.17e-02 *

Table 8: Statistical significance testing across standard datasets. Includes One-way ANOVA followed by pairwise tests (Wilcoxon/Mann-Whitney for non-normal distributions, T-test otherwise). Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns (not significant).

7 Additional Discussion

7.1 Vienna 4x22: A Stress Test for Alignment

The dramatic 0.22 F1 drop on Vienna under corruption, shared by all four methods, deserves separate examination. Vienna 4x22 [1] contains only 4 unique compositions, each performed by 22 pianists. The limited score-level diversity means that test pieces are structurally very similar to training pieces in the

clean setting; under corruption, the room for the alignment to use score-level disambiguation cues collapses, and every method drops to the ~ 0.77 F1 range. This is a corpus-level property of Vienna, not a method-specific weakness: the closest baseline (DualDTW) shows comparable degradation. We hypothesize that performance on Vienna corrupted is bounded by the maximum amount of structural information that can be inferred from heavily-perturbed performances of compositionally homogeneous works.

References

- [1] Werner Goebel, “The Vienna 4x22 Piano Corpus,” 1999. Available: http://repo.mdw.ac.at/projects/IWK/the_vienna_4x22_piano_corpus/index.html doi: 10.21939/4X22.